BIO/CSC 243 Bioinformatics
Translating Pseudocode: IF statements

Comparing two things:

Pseudocode	*Python*
☑ • if ______	• `if`
☑ • otherwise	• `else`
• ___ is bigger than ___	• `>`
• ___ is smaller than ___	• `<`
• ___ is bigger than or equal to ___	• `>=`
• ___ is smaller than or equal to ___	• `<=`
• ___ is equal to___	• `==`
• ___ is not equal to ___	• `!=`
• and	• `and`
• or	• `or`
• it is not the case that ______	• `not`

if contents of storage1 is bigger than contents of storage2

```
if storage1 > storage2:
```

if contents of storage0 is bigger than 5 and is smaller than or equal to 10

```
if storage0 > 5 and storage0 <= 10:
```

```
put TRUE into storage0
put FALSE into storage1
if contents of storage2 is equal to
contents of storage3
        return storage0
otherwise
        return storage1
```

```
if storage2 == storage3:
   return True
else:
   return False
```

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Translating Pseudocode: memory locations

Things you can do with storage locations (variables in memory)

Pseudocode	Python
• contents of storage0	• `storage0`
☑ • put "ATCG" into storage1	• `storage1 = "ATCG"`
• character #0 of storage1 (the "A")	• `storage1[0]`
• total number of characters in storage1	• `len(storage1)`
☑ • return storage0	• `return storage0`
put 34 into storage1	`storage1 = 34`
put contents of storage0 into storage1	`storage1 = storage0`
put contents of storage0 + 1 into storage0 (increment)	`storage0 = storage0 + 1`

Repetition

- ☑ • repeat the following ### lines until a counter which starts at ### reaches ### when counting by ###
- ☑ • repeat the following ### lines as long as _______

```
## examples
put a random number into storage1
put a random number into storage2
```

Return sum

```
put contents of storage1 + contents of storage2 into storage3
return storage3
```

Return "TRUE" if the two numbers are the same. If they are not, return "FALSE"

```
if contents of storage1 is equal to contents of storage2
	put "TRUE" into storage3
	return storage3
otherwise
	put "FALSE" into storage3
	return storage3
```

Increment number in storage1 (don't return an answer)

```
put contents of storage1 + 1 into storage1
```

```
## return the bigger number
if contents of storage1 is bigger than contents of storage2
	return storage1
otherwise
	return storage2

## return "YES" if thymine exists in DNA strand in storage1, else "NO"
put 0 into storage3
repeat the following 5 lines as long as contents of storage3 is smaller than total number of characters
	if character #(contents of storage3) of contents of storage1 is equal to "T"
		put "YES" into storage2
		return storage2
	otherwise
		put (1+ storage3) into storage3
put "NO" into storage2
return storage2
```